The Missing Memory Hierarchy: Demand Paging for LLM Context Windows

Tony Mason
University of British Columbia
Canada
Georgia Institute of Technology
USA
fsgeek@cs.ubc.ca
fsgeek@gatech.edu
fsgeek@wamason.com

Abstract

The context window of a large language model is not memory. It is L1 cache—a small, fast, expensive resource that the field treats as the entire memory system. There is no L2, no virtual memory, no paging. Every tool definition, every system prompt, and every stale tool result occupies context for the lifetime of the session. The result is measurable: across 857 production sessions and 4.45 billion effective input tokens, 21.8% is structural waste.

We present Pichay, a demand paging system for LLM context windows. Implemented as a transparent proxy between client and inference API, Pichay interposes on the message stream to evict stale content, detect page faults when the model re-requests evicted material, and pin working-set pages identified by fault history. In offline replay across 1.4 million simulated evictions, the fault rate is 0.0254%. In live production deployment over 681 turns, the system reduces context consumption by up to 93% (5,038KB → 339KB); under extreme sustained pressure, the system remains operational but exhibits the expected thrashing pathology, with repeated fault-in of evicted content.

The key observation is that the problems the field faces—context limits, attention degradation, cost scaling, lost state across sessions—are virtual memory problems wearing different clothes. The solutions exist: working set theory [3], demand paging, fault-driven replacement policies, and memory hierarchies with multiple eviction-managed levels. We describe the architecture of a full memory hierarchy for LLM systems (L1 through persistent storage), report on the first three levels deployed in production use (L1 eviction, L2 fault-driven pinning, L3 model-initiated conversation compaction), and identify cross-session memory as the remaining frontier.

CCS Concepts: • **Software and its engineering** → **Allocation / deallocation strategies**; **Virtual memory**; • **Computing methodologies** → *Natural language processing*.

SOSP '17, Shanghai, China
2017.

Keywords: context windows, virtual memory, demand paging, large language models, agentic AI, working set, eviction policy

ACM Reference Format:
Tony Mason. 2017. The Missing Memory Hierarchy: Demand Paging for LLM Context Windows. In *Proceedings of ACM SIGOPS 31st Symposium on Operating Systems Principles (SOSP '17).* ACM, New York, NY, USA, 16 pages.

1 Introduction

In 1961, the Atlas computer introduced demand paging: programs could address more memory than physically existed, and the hardware would transparently load pages from backing store as needed [9]. Before Atlas, programmers manually managed overlays—explicitly swapping code segments in and out of a fixed address space. The transition from overlays to virtual memory is one of the foundational advances in systems architecture.

Large language model context windows are in the overlay era. Every agentic AI tool—Claude Code, Cursor, GitHub Copilot, Windsurf—manually assembles context on every API call: tool definitions, system prompts, and the complete message history. There is no eviction policy, no demand loading, no working set estimation. The default behavior is accumulation until the context limit forces a crisis.

This paper makes two contributions. First, we provide the empirical evidence that context management is a memory management problem. We instrument 857 production sessions (54,170 API calls, 4.45 billion effective input tokens) and find that 21.8% of tokens are structural waste from three sources: unused tool schemas (11.0%), duplicated content (2.2%), and stale tool results (8.7%) reprocessed at a median amplification of 84.4×. These are not implementation bugs. They are the inevitable consequence of managing a finite resource (the context window) without the abstractions that operating systems developed for the analogous problem fifty years ago.

Second, we build the system. Pichay is a demand paging system for LLM context windows, implemented as a transparent proxy between client and inference API. It distinguishes

garbage collection (removing ephemeral tool output that cannot be re-requested) from paging (evicting addressable content that can be faulted back in). It detects page faults when the model re-requests evicted content. It pins working-set pages identified by fault history. And the eviction summaries it generates—designed as space-saving markers—function as retrieval handles that models understand without instruction: "`[Paged out: Read file.py (8,192 bytes). Re-read if needed.]`"

In production deployment, the system extends a session from 7% free context to 43% free—the difference between imminent context death and substantial remaining capacity. Over a 681-turn session, it sustains operation at 97% eviction rate, though this extreme case reveals the thrashing pathology well-known from OS research [3]: when the working set exceeds the resident set, the system spends more effort on faulting than on useful work.

The deeper observation is that the context window is not memory. It is L1 cache. And nobody is building the rest of the hierarchy. The field's response to context limits is to make L1 bigger—1M token windows, 10M token windows. This is the equivalent of building machines with more physical RAM instead of inventing virtual memory. It works, but it doesn't scale, and it misses the architectural insight: a managed hierarchy of small-fast-expensive and large-slow-cheap storage, connected by eviction policies and fault mechanisms, outperforms any single level no matter how large.

We describe the full hierarchy: L1 (the generation window), L2 (the working set, demand-paged and pinned), L3 (session history, compressed with declared losses), and L4 (cross-session persistent memory, indexed and retrievable). The first two levels are deployed and evaluated. L3 is implemented—the model can compress conversation turns into outcome summaries via cooperative cleanup tags—but not yet evaluated at scale. L4 is designed and its mechanisms identified.

The contributions:

1. A large-scale empirical characterization of context window utilization in a production agentic AI corpus, establishing that 21.8% of tokens are structural waste with a measured taxonomy.
2. Pichay, a demand paging system for LLM context windows with empirically measured fault rate (0.0254% offline, deployed in production).
3. Fault-driven pinning: a page replacement policy that learns from its own mistakes—one fault pins a page for the session, reducing repeated faults in steady-state workloads.
4. Cooperative memory management via two side channels (phantom tools and cleanup tags): mechanisms enabling the model to voluntarily release cold pages, request cached faults, and compress conversation history—a new point in the design space not available in hardware memory hierarchies, where applications are non-cooperative.
5. The architectural observation that LLM context management maps structurally (not merely metaphorically) to virtual memory, and that the full OS memory hierarchy—from cache replacement policies to working set theory—applies directly.
6. Design of a four-level memory hierarchy for LLM systems, with the first three levels deployed (L1 eviction, L2 pinning, L3 cooperative compaction) and the first two evaluated.

2 Background

2.1 Agentic AI Architecture

Modern AI coding assistants operate as agentic systems: the model receives a set of tool definitions (JSON schemas describing available operations), a system prompt (identity, instructions, behavioral constraints), and a growing message history (alternating user and assistant turns including tool invocations and their results).

On each API call, the client application assembles the full context: system prompt, tool definitions, and complete message history. The inference provider tokenizes this context, computes attention over all tokens, and generates a response. The response is appended to the message history, and the cycle repeats.

The critical property: *context is reassembled from scratch on every API call.* There is no persistent state at the inference layer between calls. The entire conversation—including tool definitions that haven't changed and results that will never be referenced again—is sent, tokenized, and attended to on every turn.

2.2 Context Windows as Physical Memory

The analogy between context windows and physical memory is structural, not merely metaphorical (Table 1).

Table 1. Virtual memory analogy for context window management.

OS Concept	Context Window Equivalent
Physical memory	Context window (200K tokens)
Virtual memory	Persistent state (disk, databases)
Page table	Retrieval handles (UUIDs, anchors)
Page fault	Model re-requests evicted content
MMU	Proxy layer between client and API
Working set	Currently relevant context subset
Demand paging	Load tool definitions on first use
Thrashing	Evict/fault cycle exceeding useful work
Pinning	Fault history prevents re-eviction

Early computer systems managed limited physical memory through manual overlays: programmers explicitly swapped code segments in and out of a fixed address space. The transition to virtual memory—Atlas [9], Multics, then Unix—automated this management through hardware page tables, demand paging, and working-set-based eviction policies [3, 4].

Context windows are in the overlay era. Applications manually manage what fits, with no system-level support for eviction, demand loading, or working set estimation.

2.3 The Memory Hierarchy

The field's response to context pressure is to increase context window size—from 4K to 32K to 128K to 1M tokens and beyond. This is the equivalent of building machines with more physical RAM instead of inventing virtual memory. It works but does not scale: attention cost is quadratic in context length, and the $O(n^2)$ cost explosion in long sessions means that even million-token windows fill and degrade [18].

Long-context performance degradation is not solely a retrieval failure. Even with perfect retrieval of relevant evidence, model accuracy can drop substantially as irrelevant context length increases (and even when irrelevant tokens are masked) [5].

The alternative is architectural: a managed hierarchy of levels, each larger, slower, and cheaper than the one above it, connected by eviction policies and fault mechanisms (Table 2).

Table 2. Memory hierarchy for LLM systems.

Level	Contents	Eviction	Fault
L1	Generation window	—	—
L2	Working set (pinned)	Pressure-based	Re-read
L3	Session history	Age-based	Summary expansion
L4	Cross-session memory	LRU / relevance	Retrieval query
Storage	Full corpus	None	Search + ingest

L1 is the active generation window—the tokens the model attends to on the current API call. Small, fast, expensive per token. This is what everyone calls "the context window."

L2 is the working set: content the model is actively using but that need not occupy L1 on every turn. Managed by fault-driven pinning (Section 3.5). A page evicted from L1 that is immediately re-requested is promoted to L2 and kept resident until context pressure forces demotion.

L3 is session history: earlier conversation turns, completed tool interactions, superseded plans. Compressed with declared losses into structured summaries. Faultable by expanding the summary back to original content from the client's backing store.

L4 is cross-session persistent memory: authored compressions (tensors), activity records, semantic indices. Survives session death. Retrieved by graph traversal or similarity search.

Storage is the full corpus: every conversation transcript, every tool output, every document. Archived, indexed, never resident unless demanded.

Content migrates between levels based on access patterns. The hot path stays in L1. Working-set content is demand-paged between L1 and L2. Cold history compresses into L3. Cross-session knowledge persists in L4. The context limit does not disappear—it becomes the L1 cache size. The total addressable memory is unbounded. System performance is determined by hit rates at each level, not by L1 capacity alone.

This paper implements and evaluates L1 management (eviction and garbage collection) and L2 management (fault-driven pinning). L3 (rolling conversation compaction) is implemented via cooperative cleanup tags (Section 3.9) but not yet evaluated at scale. We describe the interface to L4 (persistent memory stores).

2.4 Related Work

Context compression. LLMLingua [6] and related approaches compress prompts by removing tokens predicted to be low-information. These are lossy, model-dependent operations applied to content. Our interventions are structural: we remove entire categories of waste (unused tool schemas, duplicate content, dead results) without lossy compression of the remaining content.

Agent context pruning. SWE-Pruner [16] trains a 0.6B-parameter neural skimmer for task-aware context pruning in SWE-bench agents, achieving 23–54% token reduction. Lindenbauer et al. [10] show that simple observation masking halves agent cost while matching LLM-summarization solve rates. ACON [7] provides a unified framework for history and observation compression, reducing peak usage by 26–54%.

These approaches frame the problem as prompt optimization or compression. The key difference: they optimize for benchmark scores on fixed task distributions. We measure production waste profiles and implement eviction policies with an empirically measured fault rate. Their framing is "make the prompt smaller." Ours is "manage the working set." Working-set management composes with other interventions and provides measurable fault rates.

Quadratic cost analyses. Recent work has quantified the $O(n^2)$ cost explosion in long agentic sessions where cache reads dominate [18]. Each API call reprocesses the full context; as sessions grow, the cumulative cost grows quadratically with session length. Our amplification factor measurement (Section 5.1) confirms this at production scale and our interventions directly address it.

Prompt caching. Anthropic and OpenAI offer prompt caching that avoids recomputation of static prefixes. In our corpus, 93.5% of input tokens are cache reads—caching is working. But cached tokens still occupy the context window and require attention computation for every output token generated. Caching reduces the input processing cost; it does not reduce the attention cost or the memory pressure.

Context engineering. The emerging discipline of context engineering [2, 11, 17] recognizes that managing what enters context is as important as the model itself. Mei et al. [11] survey the field comprehensively, covering retrieval, processing, and management. Anthropic's own engineering team describes practical patterns [13]. Our work provides a specific mechanism (proxy-layer interposition with eviction policies) and the empirical production data that the field currently lacks.

Prompt-level memory management. MemGPT [12] pioneered the OS virtual memory analogy: a FIFO message queue with recursive summarization at capacity, plus archival/recall databases accessible via tool calls. The model can save information but cannot explicitly release it; eviction is capacity-triggered (flush at 100%), not cost-driven. Contextual Memory Virtualisation [14] models session history as a DAG with structurally lossless trimming that strips mechanical bloat (raw tool outputs, base64, metadata) while preserving conversational content verbatim—achieving up to 86% reduction for tool-heavy sessions. Focus [15] gives the agent autonomy over compression: the LLM consolidates learnings into a persistent knowledge block and prunes raw history, inspired by slime mold exploration. SideQuest [8] teaches the model to evict stale tool outputs from the KV cache via a fine-tuned parallel reasoning thread, achieving 56–65% peak memory reduction—but eviction is irreversible and requires inference engine modifications.

These systems address the architectural problem (MemGPT's VM analogy), the structural problem (CMV's lossless trimming), the autonomy problem (Focus's model-directed compression), or the inference problem (SideQuest's KV cache eviction). None derives an economic model where prompt-token retention cost scales quadratically while fault cost scales linearly. None provides cooperative recall/release primitives informed by that cost model. None reports empirical fault rates on production sessions.

Positioning. Prior work addresses either the cost problem (quadratic analyses) or the content problem (compression, pruning, RAG) but not the structural problem: context windows are unmanaged physical memory. Our contribution is the systems abstraction—working set, eviction policy, fault rate, amplification factor—and the empirical evidence that it works on production data.

3 System Design

Pichay is a transparent HTTP proxy that interposes between an agentic AI client and the inference API. On each request, it receives the full message array assembled by the client, applies context management policies, and forwards the modified request to the inference provider. The client is unaware of the interposition.

3.1 Architecture

The proxy operates at the Messages API level. It receives JSON requests containing the system prompt, tool definitions, and message history. It returns the inference provider's response unmodified. The interposition point is the message array: the proxy can inspect, measure, modify, or replace any message before forwarding.

This design has three properties: (1) it requires no changes to the client, the model, or the inference API; (2) the client retains the full, unmodified conversation history, providing a backing store from which evicted content can be faulted back in; and (3) the proxy sees the complete request on every turn, enabling stateful management without persistent connections.

3.2 Garbage Collection vs. Paging

Not all tool results are equal. We distinguish two categories:

Garbage collection applies to ephemeral tool outputs—Bash command results, search outputs, directory listings—that have no stable identity. Once consumed, they cannot be meaningfully re-requested. Removing them is reclamation of dead content.

Paging applies to addressable content—file reads, plan documents, specifications—that have stable identity (a file path). Removing these creates a fault risk: the model may need the content again and must re-read the file.

The distinction matters for fault rate calculation. Only paging (Read evictions) can produce faults. Garbage collection cannot. Conflating them inflates the eviction denominator and deflates the apparent fault rate.

3.3 Eviction Policy

The current policy is FIFO by user-turn age: tool results older than τ user turns and larger than $s_{\min}$ bytes are candidates for eviction ($\tau = 4$, $s_{\min} = 500$ in all experiments). This is deliberately simple. The research question is how far a minimal policy extends before requiring sophistication.

Evicted content is replaced with a retrieval handle:

```
[Paged out: Read /path/to/file.py (8,192 bytes,
187 lines). Re-read the file if you need its
content.]
```

The handle serves three functions: (1) it identifies what was removed; (2) it states how to recover the content; and (3) it occupies substantially less context than the original.

3.4 Page Fault Detection

When the model issues a tool call that matches an evicted entry (same tool name and arguments), the proxy detects a *page fault*: the model is requesting content it previously had but lost to eviction. The evicted entry is indexed by a key derived from the tool name and input arguments (e.g., file path for Read operations).

Page faults are observable events. They measure the cost of eviction: each fault is a wasted tool call round-trip that would not have occurred if the content had remained resident. At inference-time pricing, faults have direct monetary cost.

3.5 Fault-Driven Pinning

The simplest upgrade to FIFO: if evicting a page caused a fault, don't evict it again. One fault per file, permanently pinned for the session. The algorithm:

1. On eviction of addressable content: record the file path and content hash.
2. On page fault: record the evicted content's hash in a fault history table.
3. On next eviction attempt for the same path: if the current content hash matches the fault history entry, pin the page—skip eviction permanently.
4. On new read of a pinned path with different content (file was edited): unpin. The old version is stale; the new version starts a fresh fault cycle.

The content hash comparison prevents false pins: if a file changed between eviction and re-read, the eviction was correct (stale data was removed). Pinning only applies when the model demonstrably needed exactly what was taken away. The unpin-on-edit rule handles the edit/review cycle common in coding sessions.

3.6 Retrieval Handles as Anchors

The eviction summary was designed as a space-saving marker. In practice, it functions as a late-binding retrieval handle—an anchor that stores minimal metadata and resolves to full content on demand.

The handle resolves to *current* content, not the original evicted content. A file edited since eviction materializes at its new state when faulted in. This temporal property is a feature: the model always gets the latest version, and stale cached content is impossible.

Behavioral evidence: when a fresh model instance resumed a session containing paged-out content, it unprompted stated: "Let me re-read the files I need since some were paged out." The model recognized the handles, understood what was missing, and chose to fault content in before acting. The handle's format carries its own semantics—no instruction was needed.

3.7 Cooperative Memory Management

In OS virtual memory, the application is non-cooperative. It never voluntarily releases pages. The OS must infer the working set from access patterns because the CPU will not say what it needs next. Decades of replacement algorithms—FIFO, LRU, Clock, Working Set—exist because of this fundamental asymmetry.

LLMs break this assumption. The model has *incentive* to cooperate with the memory manager: cleaner context means better attention, better output quality, and longer session life. The processor wants to help manage its own cache. This changes the design space.

Pichay exploits this through *phantom tools*—tool definitions injected by the proxy that the framework never sees. The model can call them; the proxy intercepts the calls from the streaming response before the framework receives them, handles them internally, and injects coherent tool results on the next turn. The framework is unaware of the side channel.

Two phantom tools are defined:

`memory_release(paths)` The model signals it no longer needs specific files. The proxy marks them for immediate eviction, bypassing the age threshold. This is the reference bit—provided voluntarily by the processor.

`memory_fault(paths)` The model requests evicted content restored from the proxy's backing store. The proxy resolves the fault from its eviction cache without a file system round trip—faster and cheaper than a real Read tool call.

The second cooperative channel is *cleanup tags*—structured directives embedded in the model's output text, parsed by the proxy before forwarding to the framework. Where phantom tools are proxy-to-model (the proxy offers capabilities), cleanup tags are model-to-proxy (the model initiates management). Four operations are defined:

`drop: block:ID` Immediately evict a specific block.

`summarize: block:ID "text"` Replace a block with a compact summary—lossy compression authored by the model, not the system.

`anchor: block:ID` Pin a block against eviction.

`collapse: turns N-M "text"` Compress a range of conversation turns into a single synthetic summary block, removing all intermediate content.

The collapse operation is particularly significant: it enables L3 compaction (Section 3.9) initiated by the model itself, based on its understanding of which dialogue was scaffolding and which produced durable outcomes.

The key insight: in hardware memory hierarchies, the application is adversarial or indifferent. The entire replacement algorithm literature assumes non-cooperation. Cooperative demand paging—where the processor voluntarily releases cold pages and explicitly requests faults—is a new point in the design space, enabled by the fact that LLMs experience degraded output quality under context pressure and can

learn that cooperation improves their own performance. The two channels (phantom tools and cleanup tags) provide both directions of the cooperative protocol: the proxy informs the model of memory state, and the model directs the proxy to act on its cognitive assessment of what is cold.

3.8 Graduated Pressure Zones

The eviction policy must respond to context pressure, but a binary threshold (evict/don't-evict) is too coarse. Pichay defines four pressure zones based on token consumption:

Normal (< 60K tokens): No intervention. The proxy observes and logs.

Advisory (60K–100K): The proxy injects memory pressure information into the model's context—current fill percentage, the five largest resident blocks, and available cleanup operations. The model can act cooperatively or ignore the advisory. This is the graduated equivalent of the "low memory" notification in desktop operating systems.

Involuntary (100K–120K): The proxy begins automatic eviction using the FIFO age policy. The model is informed but not consulted.

Aggressive (≥ 120K): Emergency eviction with relaxed thresholds. Context survival takes priority over working set preservation.

The advisory zone is the cooperative innovation. Rather than evicting silently (the OS approach) or crashing at capacity (the current agentic AI approach), the system provides the model with enough information to make intelligent cleanup decisions. The 60K threshold was chosen to provide approximately 40K tokens of runway before involuntary eviction—enough for the model to complete a coherent thought and emit cleanup tags before losing agency over the process.

3.9 L3: Rolling Conversation Compaction

The proxy manages tool results (L1/L2). Non-tool content—conversation history, planning dialogue, orientation reads—is managed by the collapse operation (Section 3.7). This is L3 in the memory hierarchy: session history compressed with declared losses into structured summaries.

The collapse operation `collapse: turns N-M "summary"` replaces all blocks in a contiguous turn range with a single synthetic block containing the model-authored summary. The original content is not stored—this is lossy compression by design. The summary captures *outcomes* (what was decided, what was built, what failed) rather than *process* (which files were read, what intermediate steps occurred).

Block state persists across session restarts via atomic checkpointing. On each cleanup tag processing pass, the proxy serializes block metadata (IDs, content hashes, sizes, turns, roles, status, summaries) to a JSON checkpoint file using atomic write (tmp file + rename). On session creation, the checkpoint is loaded and block tracking resumes from the persisted state. Original content is lazily repopulated as the message array is reprocessed.

The checkpoint is metadata-only—content is not persisted (the client's message array is the backing store). This keeps checkpoint files small (kilobytes, not megabytes) and avoids the consistency hazard of maintaining two copies of message content.

4 Measurement

4.1 Corpus

We instrument Claude Code, Anthropic's AI coding assistant, across a single power user's sessions over approximately four months (November 2025 through March 2026). The corpus comprises 857 sessions across 15 software projects, drawn from two sources: a local WSL environment and an Ubuntu VM (migrated from a prior machine).

Sessions are classified by type:

- **Main sessions** (59): Human-facing conversations where the user interacts directly with the assistant.
- **Subagent sessions** (567): Delegated tasks spawned by the main session to handle subtasks in parallel.
- **Compact sessions** (154): Context compaction continuations triggered when the main session approaches its context limit (the client automatically summarizes conversation history to stay within the context window).
- **Prompt suggestion sessions** (21): Short sessions for generating prompt suggestions.

The corpus represents 54,170 API calls and 4.45 billion effective input tokens (input tokens + cache creation + cache read tokens). Token usage is extracted from the API response metadata embedded in session transcripts.

Because the live corpora continue to evolve, the paper reports a frozen cohort (857 sessions) rather than a moving live count. We provide a recount script (`tools/corpus_counts.py`) to make the counting boundary and dedup assumptions explicit. Unless stated otherwise, all corpus tables and figures in this draft use the frozen cohort with a fixed counting protocol (`dedup_by_content`, `min_size=10000`) so results are stable across revisions.

Bias declaration. This is a single-user corpus from a power user who regularly exhausts context windows across complex, multi-file engineering tasks. We argue this is representative of the class where context management matters most, and note that Pareto-distributed usage means this class likely dominates total fleet token consumption. Multi-user validation is identified as future work.

4.2 Instruments

We develop three instruments at different layers of the stack:

Probe. (`probe.py`): A streaming JSONL analyzer that reads Claude Code's raw session transcripts. Classifies records by

type (user, assistant, tool_result, progress), measures content sizes, tracks tool usage, and computes per-session metrics including amplification factor and tool overhead ratio. Requires no API calls; operates on existing session files.

Proxy. (`proxy.py`): A transparent HTTP proxy deployed between Claude Code and the Anthropic API. Intercepts every request, capturing system prompts, tool definitions, message arrays, and token usage. Optionally applies interventions (eviction, trimming) before forwarding. Logs all decisions to JSONL for offline analysis. This is the "MMU" in our analogy.

Experiment runner. (`pichay`): An experiment framework that orchestrates paired runs under different treatment conditions (baseline, trimmed, compact+trim), captures all artifacts (proxy logs, session data, git state, test results), and computes comparative metrics. The framework is self-bootstrapping: it was built by Claude Code running through its own proxy, with each round's logs feeding the next round's analysis.

4.3 Treatment Conditions

We define three treatment conditions applied at the proxy layer:

1. **Baseline**: Proxy in observation mode. Logs all traffic without mutation.
2. **Trimmed**: Tool definition stubbing and skill deduplication applied to each request before forwarding.
3. **Compact+Trim**: Stale result eviction (paging) combined with trimming.

All treatments use the same model (Claude Opus), same temperature, and same task prompts. The proxy's interposition is transparent to both the client and the API—neither is aware of the mutations.

5 Results

The analysis proceeds in two phases. Phase 1 characterizes broad conversational patterns—tool overhead ratios, amplification factors, and token accounting—across the full 857-session corpus using session transcript analysis. Phase 2 performs detailed API-level waste decomposition on a smaller proxy-captured sample (5 sessions, 99 API calls) where we have full request payloads. The corpus-scale projection (Section 5.6) then bridges the two phases, applying measured conversion constants from Phase 2 to the full corpus to estimate fleet-wide addressable waste. The conversion constant used for projection (bytes-to-token ratio) is estimated from a broader proxy sample of 139 API calls.

5.1 Phase 1: Conversation-Level Waste

Tool overhead. Across the full corpus, 79.4% of conversation bytes are tool results. Assistant text accounts for 12.7% and user text for 7.9%. Two independent measurements converge on this ratio (78.2% and 79.4%), suggesting it is a stable property of the workload, not an artifact of sampling.

Tool type concentration. Tool usage is extremely concentrated (Figure 1). `Read` accounts for 75% of all tool output bytes (9,393 calls, mean 7,935 bytes per result). `Bash` accounts for 13.3% (10,090 calls). All other tools contribute less than 5% each. This concentration has a direct implication: file content read by the `Read` tool is the dominant source of context bloat.

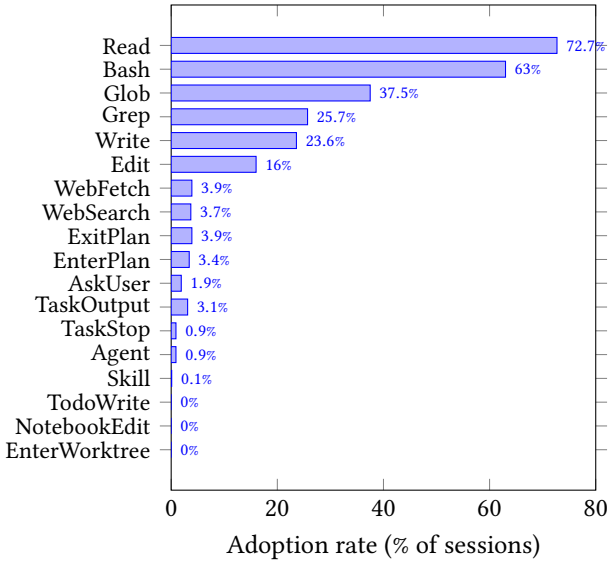

Figure 1. Tool adoption rate across 801 active sessions. The median session uses 3 of 18 available tools. Seven tools see zero or near-zero adoption, yet their full schemas (collectively ~24,500 bytes) are sent on every API call.

Amplification factor. Each tool result persists in context from the turn it was generated until the session ends (or the client's internal compaction evicts it). We define the *amplification factor* as:

$$A = \frac{\sum_{r \in R} \text{size}(r) \times \text{turns_survived}(r)}{\sum_{r \in R} \text{size}(r)}$$

where R is the set of tool results and turns_survived is the number of subsequent turns the result remains in context. This measures how many times, on average, each byte of tool output is reprocessed.

For main sessions: median $A = 84.4\times$, P75 = 217.9×, P90 = 570.8×. For subagents: median $A = 12.8\times$ (short-lived, less accumulation). Amplification scales linearly with session length at a ratio of approximately 0.5, confirming the quadratic cost structure identified in recent analyses [18].

Token accounting. The corpus consumed 4.45 billion effective input tokens with a 93.5% cache hit ratio, meaning the vast majority of input processing is served from cache. However, cached tokens still occupy the context window and require attention computation for every output token. The average API call processes 82,061 effective input tokens and generates 88 output tokens—a 933:1 input-to-output ratio. Agentic coding is an overwhelmingly input-bound workload.

5.2 Phase 2: API-Level Waste Taxonomy

Using the proxy, we decompose 5 sessions (99 API calls, 24.4 MB of request data) into four categories of addressable waste (Table 3).

Table 3. Waste taxonomy. Percentages are of total request bytes across 99 API calls.

Category	Bytes	%	Mechanism
Dead tool output	6,468,360	26.5	Stale results never re-referenced
Tool definition stubs	4,924,950	20.2	Schemas for unused tools
Static re-send	2,680,794	11.0	Unchanged system prompt content
Skill triplication	700,582	2.9	Same skill listed 3×
Total addressable	**14,774,686**	**60.5**	

Dead tool output (26.5%). Tool results from early turns persist in context long after any reference to them. A file read during orientation (Q1 of the session) survives approximately 90% of the remaining session. The content is reprocessed on every subsequent API call, consuming attention without contributing to the current task.

Tool definition schemas (20.2%). Claude Code sends 18 tool definitions totaling 63,088 bytes on every API call. The median session uses 3 tools. The remaining 15 tool schemas (~52,500 bytes) are sent, tokenized, and attended to on every call without ever being invoked.

Static system content (11.0%). The system prompt, CLAUDE.md instructions, and memory files are re-sent identically on every API call. While prompt caching avoids the forward-pass cost, these tokens still occupy the context window and participate in attention computation.

Skill triplication (2.9%). The skills list—describing available slash commands—is injected into messages three times under different prefixes (`base`, `example-skills:base`, `document-skills:base`). Simple deduplication removes two-thirds of the entries.

5.3 Interventions

All interventions operate in the proxy layer between the client application and the inference API. No changes to the model, the client, or the API are required.

Tool definition stubbing. Unused tool definitions are replaced with minimal stubs:

```
{"name": "NotebookEdit",
 "description": "...(first line)...",
 "input_schema": {"type":"object","properties":{}}}
```

Full schemas (~3,505 bytes each) are replaced with stubs (~80 bytes). On first invocation of a stubbed tool, the full definition is restored from a stored copy. The intervention is session-scoped: once a tool is used, its schema remains restored for the session. Per-request savings: $(18 - k) \times 3{,}425$ bytes, where k is the number of tools used so far.

Content deduplication. Skill entries are parsed, grouped by base name, and deduplicated (keeping the first occurrence). Saves 7,453 bytes per request. Static system prompt components are tracked by content hash across turns; identical components are logged as candidates for prefix caching. Currently measurement-only for static components—actual stripping requires cache-aware API support.

Stale result eviction (paging). Tool results older than a threshold (4 user-turns from the end of the conversation) and larger than a minimum size (500 bytes) are evicted. Error results are never evicted (the model needs them for debugging). Evicted content is replaced with a summary:

```
[Paged out: Read /path/to/file.py (12,450
bytes, 287 lines). Re-read if needed.]
```

The summary preserves the tool name, key parameter, and original size (~200 bytes regardless of original size). If the model re-invokes the same tool with the same parameters, this constitutes a *page fault*—the eviction was incorrect and the content was still needed.

5.4 Eviction Safety

We validate eviction safety through offline replay: 29 proxy-captured sessions are replayed through the pager, simulating eviction decisions without making API calls.

Table 4. Eviction safety results from offline replay.

Total simulated evictions	1,393,000
Page faults detected	354
Fault rate	**0.0254%**
Content evicted	8.49 GB

A fault rate of 0.0254% means the eviction policy is sound: content older than 4 user-turns is almost never needed again. The 354 faults represent content that was genuinely dead by the age criterion but happened to match a later request. A fault rate of zero would indicate over-conservative eviction; some faults are expected and acceptable.

Table 5. Treatment comparison on a standardized task.

Metric	Baseline	Trimmed	Compact+Trim
API calls	3	4	3
Effective input tokens	114,222	88,421	71,816
Cache read tokens	79,712	38,639	32,228
Task completed correctly	Yes	Yes	Yes
Token reduction	—	22.6%	**37.1%**

5.5 Live Treatment Comparison

We run a standardized task under three treatment conditions (Table 5).

The compact+trim treatment achieves 37.1% reduction in effective input tokens. Cache reads drop 59.6%. The task completes correctly under all conditions in this controlled run.

5.6 Corpus-Scale Projection

We validate the small-sample findings across the full 857-session corpus using measured constants from the proxy sessions and tool usage patterns extracted from raw session transcripts.

The bytes-to-token conversion ratio of 4.15 bytes per effective input token is measured from 139 proxy-captured API calls.

Table 6. Corpus-scale token savings projection.

Intervention	Tokens Saved	% of Input
Tool stub trimming	487.5M	11.0%
Skill deduplication	95.8M	2.2%
Static re-send	387.0M	8.7%
Total addressable	**970.4M**	**21.8%**

The average saving is 17,913 tokens per API call. Across the corpus, this translates to 85 billion fewer token-token attention pairs (17,913 context tokens × 88 output tokens × 54,170 calls).

5.7 Production Deployment

We deploy Pichay in compact mode as the proxy for our own development workflow—the system is used daily by the authors for software engineering tasks. This section reports on two representative sessions that illustrate different operating regimes.

Session A: Steady-state coding. A standard coding session using compact mode ($\tau = 4$, $s_{\min} = 500$):

The session recovered 36 percentage points of context capacity—the difference between imminent context exhaustion and substantial remaining headroom. Of 15 evictions, 11 were garbage collection (Bash/Grep/Glob outputs) and 4 were Read evictions. The single fault was a plan file read early in the session and evicted when it crossed the age threshold. The plan file is reference material needed for the entire session—a hot page that FIFO treats as cold. This is the classic working set failure: the eviction policy measures age, not access pattern.

Table 7. Production Session A: steady-state coding.

Context free before compaction	7%
Context free after compaction	43%
Total evictions	15
Garbage collected	11
Read evictions (pageable)	4
Page faults	1
Fault rate (Read only)	25%

Session B: Sustained multi-agent coordination. A 681-turn session involving multiple subagent spawns and cross-project file analysis:

Table 8. Production Session B: 681-turn sustained session.

Turns	681
Evictions (total)	680
Garbage collected	74
Read evictions (pageable)	606
Page faults	659
Fault rate (total evictions)	**97% (659/680)**
Peak compression	5,038KB → 339KB
Session termination	API rate limit

The 97% fault rate is a pathology, not a feature. The system evicted almost everything and the model re-requested almost everything. Three specific patterns emerged:

Thrashing cycle. Three files (3,438 + 1,570 + 1,592 bytes) were evicted and faulted in a cycle across turns 163–170+. The working set exceeded the resident set—classic thrashing [3].

Sequential scan. During a planning phase, the model read across three code repositories, cycling through the same 7 files repeatedly. The working set for planning was larger than what the age threshold allowed to remain resident.

Self-inflicted inflation. The 5,038KB pre-compaction size includes re-reads caused by previous evictions. Without the eviction-fault cycle, the original content would have been substantially smaller. The proxy was measuring its own overhead as "bytes saved"—a metric artifact of thrashing.

Cost. The session terminated by hitting the API rate limit, not the context limit. At 659 faults (each an inference-priced tool call round-trip), thrashing consumed rate budget faster than useful work. This demonstrates that fault cost is not merely computational but monetary.

Anchor handle behavior. When a fresh model instance resumed a session containing paged-out content, it stated: "Let me re-read the files I need since some were paged out, then create the task list and start implementing." The model recognized the retrieval handles, understood what was missing, and chose to fault content in before acting—without any instruction to do so. The handle format is self-describing: the model infers the recovery mechanism from the text of the summary alone.

5.8 Cognitive Transaction Protocol Evaluation

In a separate experiment track, we evaluate whether a model can operate as a bounded cognitive transaction engine under an explicit validator/reducer contract. The paper-ready tables are generated from versioned artifacts under `experiments/cognitive_transactions/synthesis/`.

Phase 2: Candidate viability under protocol induction. Table 9 reports per-model validity with and without synthetic teacher-trace guidance.

Table 9. Cognitive transactions Phase 2 (3 replicates, 15 steps per condition).

Model	Valid % (none)	Valid % (synthetic)
inception/mercury-2	6.7	100.0
meta-llama/llama-4-maverick	0.0	100.0
allenai/olmo-3.1-32b-instruct	0.0	100.0
moonshotai/kimi-k2.5	0.0	73.3

The main result is an induction gap: under synthetic guidance, three of four candidates reach 100% valid transitions; without guidance, three of four collapse to 0% validity.

Phase 3: Guidance ablation and contract tightening. Table 10 shows that reduced guidance causes an abrupt drop for OLMo-3 (86.7% → 6.7%), while focused contract tightening recovers legality to 100% in a larger retest.

Table 10. Cognitive transactions Phase 3 (OLMo-3).

Condition	Steps	Valid	Valid %	Req-op misses
synthetic	15	13	86.7	0
synthetic_reduced	15	1	6.7	6
synthetic_followup	25	23	92.0	0
synthetic_followup_tweak	50	50	100.0	0

Phase 4a: Realism and completion-budget interaction. Table 11 reports the corrected realism run (`phase4_20260311T173018Z`). No truncation or content-filter events occurred in this run. The 2000 vs. 4000 setting is the per-step completion token cap (`max_tokens`), not the model context window. Under this setting, task-dependent variations emerge: 4000 improves failure recovery, while 2000 performs better on long-horizon durability and multitask contradiction reconciliation.

Table 11. Cognitive transactions Phase 4: working-set realism under induced handle recall (3 replicates per task/profile).

Task	Valid % @2000	Valid % @4000	Delta (2000-40
failure_recovery	80.0	100.0	-2
long_horizon_durability	86.7	83.3	+
multitask_benchmark	72.2	66.7	+

This section supports a bounded claim: capability is real under protocol scaffolding, but multitask contradiction handling remains the current boundary and should be reported as a limitation.

A confirmatory rerun on 2026-03-11 (`phase4_20260311T181454Z`) demonstrates run-to-run variance in cognitive transaction success while preserving the same qualitative boundary. Relative to the corrected baseline (`phase4_20260311T173018Z`), validity shifted by −20.0 points for failure_recovery@4000 (100.0% → 80.0%) and +5.6 points for multitask_benchmark@4000 (66.7% → 72.2%), with truncation/content-filter totals remaining zero across all task/profile pairs.

Phase 4b: Working-set realism via handle recall. Separately, we implement a working-set realism mode where evicted handles have explicit payload identity (hash, size) and `memory_fault` restores the full payload into the next-step input state. This enables controlled working-set sweeps by varying evicted handle payload size (`handle_bytes`) while keeping completion caps high to avoid structured-output truncation. A representative run with `handle_bytes=5000` (`phase4_20260311T194940Z`) reports non-trivial recall volume alongside task success, making working-set usage auditable in tokens and bytes.

6 Discussion

6.1 Why This Waste Exists

The append-only pattern in agentic AI tools is not an implementation bug. It arises from four reinforcing factors:

1. **Training data**: Models learn from conversations that grow monotonically. No training examples demonstrate intelligent content removal.
2. **API design**: The Messages API accepts a list of messages. The natural operation on a list is append. The API provides no mechanism for "this content is stale."
3. **Framework defaults**: Every orchestration framework appends by default. Eviction requires explicit engineering.
4. **Invisible cost**: Token consumption is billed after the fact. Quality degradation from context bloat is invisible unless measured. There is no backpressure signal.

6.2 The Inverted Cost Model

In traditional virtual memory, keeping a page in physical memory is free. RAM is allocated whether the page is accessed or not; the only cost is *faulting*—the latency of disk I/O when a needed page is not resident. The entire replacement algorithm literature, from FIFO to LRU to Clock to Working Set, optimizes a single objective: *minimize faults*.

In LLM context management, the cost model is inverted. Every token kept in context costs money on every turn. A 5,000-token file sitting in context for 20 turns costs 100,000 input tokens of processing. Re-reading that file once when actually needed costs 5,000 tokens. The eviction saves 95,000 tokens. Keeping is expensive; faulting is cheap.

This inversion changes the optimization objective. The correct formulation is not "minimize faults" but:

$$\min \sum_{p \in P} \left[C_{\text{keep}}(p) + C_{\text{fault}}(p) \right]$$

where $C_{\text{keep}}(p) = |p| \cdot T_{\text{resident}}(p) \cdot c_{\text{token}}$ is the cumulative cost of keeping page p resident for T_{resident} turns, and $C_{\text{fault}}(p) = |p| \cdot c_{\text{token}}$ is the one-time cost of restoring the page (one turn of reprocessing). The break-even condition is:

$$|p| \cdot T_{\text{until_next_ref}}(p) \cdot c_{\text{token}} > |p| \cdot c_{\text{token}}$$

which simplifies to: *evict whenever the page will not be referenced for more than one turn*. This is why FIFO works so well in our system despite being the worst-performing policy in classical VM: when keeping is expensive and faulting is cheap, aggressive eviction is correct by default. Sophistication in replacement policy is needed only to avoid evicting pages that will be needed on the *very next* turn.

Belady's MIN under inverted costs. Belady's optimal algorithm [1] evicts the page whose next reference is farthest in the future, minimizing total faults. Under the inverted cost model, the optimal policy is different: it must minimize the sum of keeping costs and fault costs. A page with a distant next reference should be evicted immediately (saving many turns of keep cost at the price of one fault). A page referenced every turn should never be evicted (the fault cost equals one turn of keep cost, so eviction saves nothing). The optimal offline policy under inverted costs is not MIN—it is a cost-weighted variant where pages are evicted when their projected keep cost exceeds their fault cost, regardless of whether a fault will occur.

This has a practical implication: the *size* of the page matters for eviction priority in a way it does not in classical VM (where all pages are equal size). A 10,000-token file costs ten times as much per turn as a 1,000-token file. Large pages should be evicted eagerly unless access frequency justifies the keep cost.

Non-linear fault cost. The linear fault cost $C_{\text{fault}}(p) = |p| \cdot c_{\text{token}}$ above is a simplification. Transformer inference is dominated by self-attention, which is $O(n^2)$ in sequence length. A page fault requires an additional full inference pass—the model emits a `tool_use` for the recall handle, and the restored content returns as a `tool_result` message that triggers a second inference over the entire context. The true fault cost is therefore proportional to n^2 at the current context size n, not to the page size $|p|$ alone.

This produces a counter-intuitive policy gradient. At low fill (e.g., $n = 40$K), faults are cheap and aggressive eviction is correct—the quadratic cost of an extra pass is modest, and keeping the working set small reduces the n^2 base for all subsequent turns. At high fill (e.g., $n > 100$K), faults become expensive: the additional inference pass costs roughly $(n + |p|)^2 \approx n^2$ tokens of compute. The eviction policy should therefore become *more conservative* as context pressure rises—the opposite of the naive instinct to evict aggressively under pressure. At high fill, the system should evict only content it is highly confident will not be referenced again.

Object-based cost variance. Unlike classical VM, where all pages are fixed-size blocks (typically 4 KB), Pichay manages variable-size objects: a recalled tensor may be a 3-line summary or a 200-line file. This means eviction and fault costs vary by orders of magnitude across the working set. The replacement policy cannot treat all eviction candidates as equivalent; it must weight both the per-turn keep cost and the potential fault cost by object size. Combined with the quadratic fault penalty, this argues for a *size-aware, fill-sensitive* replacement policy that has no direct analogue in the classical VM literature.

Cache invalidation cost. The cost model so far considers only token-level costs (keeping and faulting). Structural mutations—collapse operations that remove or replace blocks in the message array—have an additional cost: *prompt cache invalidation*. Inference providers cache the tokenized prefix of repeated requests; when the prefix changes, the cache misses and the entire context must be reprocessed.

In production, a collapse operation that compressed 12 turns of orientation dialogue into a single summary sentence caused the cache hit rate to drop from 100% to 25% for one turn, then recover to 100% on the following turn as the new prefix stabilized. The cost was one full recompute of ~105K tokens—comparable to the cost of several page faults. A collapse that saves 10KB of context but invalidates a 100K-token cached prefix is a net loss unless the space savings persist for enough subsequent turns to amortize the one-time recompute.

This argues for *batching* structural mutations: accumulate collapse candidates and execute them together in a single pass, paying the cache invalidation cost once rather than per-operation. It also argues against frequent small collapses in favor of infrequent large ones.

Pin decay. The inverted cost model also argues against permanent pins. In our current design (Section 3.5), one fault

pins a page permanently. But a fault tells us the content was needed *then*, not forever. Under the cost model, a pin should decay: its strength halves every K turns since last access, and the page becomes evictable when the projected keep cost of the remaining pin lifetime exceeds the fault cost. This gives LRU-like behavior with cost-weighted decay, and prevents the monotonic working-set growth that permanent pinning causes in long sessions.

6.3 Generalizability

The waste patterns we measure are structural consequences of the agentic architecture:

- Any system that sends tool definitions on every request will send schemas for unused tools. The specific percentage depends on tool count and usage distribution, but the mechanism is universal.
- Any system that injects instructions into messages risks duplication. Claude Code's triplication is egregious but not unique.
- Any system that keeps tool results in context for the session lifetime will see amplification proportional to session length.

Our empirical evidence is from one system (Claude Code) and one user. The structural argument is strong—the architecture forces the waste—but comparative measurement of other tools would strengthen the generalizability claim.

6.4 Scope, Non-Goals, and Threats to Validity

Scope. This paper evaluates context waste and demand paging behavior for agentic coding workloads with persistent tool and transcript history. The target claim is mechanism-level: when requests repeatedly resend large static prefixes and stale tool output, eviction plus fault-based restoration reduces cumulative input processing.

Non-goals. We do not claim universal percentage reductions across all assistants, all model families, or all task domains. We do not evaluate frontier reasoning benchmarks, autonomous planning quality, or human preference at product scale. We also do not claim that L1 expansion (larger context windows) is obsolete; we claim it is incomplete without hierarchical management.

Threats to validity. External validity is limited by the single-user corpus and one primary assistant implementation. Internal validity risks include prompt drift, toolset changes across versions, and workload phase effects that alter hot-set composition. To mitigate drift, we report a frozen cohort for headline numbers and provide executable recount tooling plus timestamped live snapshot artifacts for auditability.

6.5 Quality: LLM-Judged Equivalence Check

The intervention must not degrade output quality. We therefore run a paired equivalence-style check: if results are broadly comparable while compute drops, the intervention is useful.

Protocol. We select 18 sessions from the corpus (426–1,354 messages, 340K–718K characters of context). For each session, we construct paired contexts at 65–75% of the conversation: the *baseline* retains all messages; the *treatment* replaces consumed tool results outside a 20-message recency window with tombstones, modeling Pichay's eviction behavior. Both conditions receive the same continuation prompt (the next user message). Outputs are generated by Sonnet 4 under both conditions.

An ensemble of three LLM judges (one Sonnet 4, two Haiku 4.5) evaluates each paired output on correctness, completeness, and coherence (1–5 scale). Judges also state a blind preference (A, B, or tie) and whether they can identify which output had reduced context. A/B assignment is randomized per judge to prevent position bias; scores are remapped to baseline/treatment before aggregation.

Table 12. Non-inferiority evaluation: 18 sessions, 54 verdicts. Treatment evicts consumed tool results outside a 20-message recency window (mean compression 48%).

Metric	Baseline	Treatment
Judges preferring	15 (28%)	20 (37%)
Ties	19 (35%)	
Mean correctness	3.89	3.74
Mean completeness	3.59	3.59
Mean coherence	3.74	3.69
Max score Δ	0.15	
Detection rate	57% ($p = 0.14$, n.s.)	

Results. Judges prefer the treatment *more often* than the baseline (37% vs. 28%), with 35% ties. Completeness scores are identical; correctness and coherence differ by at most 0.15 points on a 5-point scale. Detection rate (57%) is not significantly above chance (binomial $z = 1.09$, $p = 0.14$, $n = 54$).

Failure modes. Two sessions (11%) produced degenerate treatment outputs (zero characters): the continuation prompt implicitly referenced content that existed only in a tombstoned tool result. These are genuine eviction casualties. The failure pattern is specific: the user's prompt references a recently-consumed tool result by content rather than by name. A reference-aware eviction policy would detect such forward dependencies; the current heuristic (consumed + outside recency window = evictable) does not.

The attention-concentration effect. In several sessions, all three judges preferred the treatment output. Removing half the context did not merely preserve quality—it improved it. This is consistent with the attention-dilution hypothesis: transformer attention distributes weight across all tokens in context; irrelevant tokens (stale tool results, consumed intermediate outputs) dilute attention on the tokens that matter. Evicting them concentrates attention on signal. The effect is strongest in sessions with high tool-result density, where the evicted content is genuinely noise. We note this as preliminary evidence; establishing the effect rigorously would require controlled attention analysis with larger N.

6.6 Compute and Energy Implications

The 970 million tokens of addressable waste in our corpus have three compute cost dimensions:

Input processing. At a 93.5% cache hit rate, most saved tokens are cache reads (907M tokens at reduced cost). However, 63M tokens are cache creation or uncached—full forward-pass computation that need not occur.

Attention cost. For every output token, the model computes attention over the full context. Removing 17,913 tokens from a mean context of 82,061 reduces the per-output-token attention computation by 21.8%. Across the corpus, this eliminates 85 billion token-token attention pairs.

Throughput. Each token in context requires KV cache memory. For large models, the per-token KV cache footprint is on the order of megabytes. Smaller context per request means less KV cache memory per request, which means more concurrent requests per GPU. A 21.8% context reduction translates, to first approximation, to 21.8% more concurrent users on the same GPU fleet.

This is the argument that changes the infrastructure economics. The savings are not just "less cost per request"—they are "more requests per machine," which is the difference between building new data centers and not building them.

Compounding across sessions. These per-call savings compound. Waste prevented on turn N is not merely saved once—it is absent from every subsequent turn's context, where it would have been reprocessed under $O(n^2)$ attention cost. The total compute savings across a session are therefore superlinear in the waste fraction and grow with session length. For a 100-turn session with 21.8% addressable waste, the cumulative attention savings are not 21.8% of total session compute—they are substantially larger, because the quadratic cost of carrying unnecessary tokens compounds on every subsequent call. The infrastructure implication follows directly: the fastest tokens are the ones you never process.

Figure 2 illustrates this compounding on an 88-turn live session running through Pichay. Cumulative baseline cost

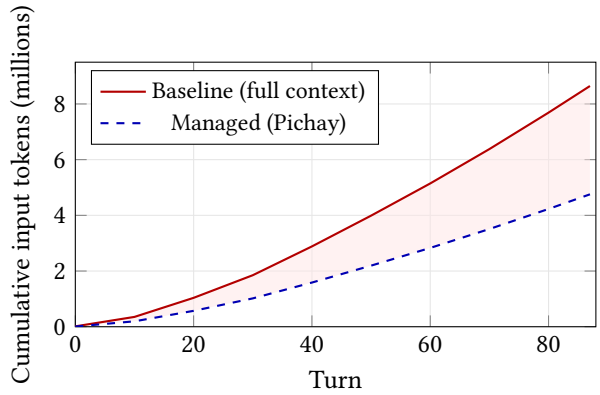

Figure 2. Cumulative input tokens processed over an 88-turn session. Without context management, every turn reprocesses all prior content, producing superlinear cumulative cost. Pichay's eviction reduces cumulative token spend by 45% ($11.67 at Opus pricing). The shaded area represents wasted computation.

(processing all context every turn) reaches 8.6M input tokens; managed cost reaches 4.8M—a 45% cumulative reduction, substantially larger than the per-turn compression ratio, because each evicted token is absent from *every subsequent turn*.

Fleet-scale extrapolation. Our corpus represents one user over four months. A fleet-wide projection requires per-user token consumption data that inference providers hold but have not published. The extrapolation framework is straightforward: N users × 970M tokens per user = total addressable waste. The mechanism (tool schema waste, content duplication, result amplification) is architectural, so the per-user waste fraction should be relatively stable even if absolute token counts vary.

7 Future Work

L3 evaluation. The collapse mechanism (Section 3.9) is implemented but not yet evaluated at scale. Open questions include: how much context does model-authored compaction recover compared to system-initiated eviction? Do declared losses in collapse summaries enable accurate fault decisions (can the model determine from a summary alone whether it needs the original content)? What is the quality impact of replacing 20 turns of dialogue scaffolding with a single outcome sentence? These require longitudinal measurement across sessions of varying length and task complexity.

Cost-aware eviction pressure. The current eviction policy uses a space-based threshold (context window capacity). The inverted cost model argues for cost-based pressure: 23% memory pressure sounds low, but 45,000 tokens per turn at inference pricing is real money. Eviction pressure should be denominated in token-turns (the product of page size and

projected remaining lifetime), not in fractional capacity. This reframes eviction from "are we running out of space?" to "is this page earning its keep?"

Cost-weighted pin decay. Pinned pages should not remain resident forever. A fault tells us the content was needed *then*, not forever. We propose cost-weighted decay: pin strength halves every K turns since last access. A page becomes evictable when its projected keep cost exceeds its fault cost. Under low memory pressure, this is permissive (pins persist). Under high pressure, cold pins are released first. This gives LRU-like behavior with cost weighting, and prevents the monotonic working-set growth that permanent pinning causes in long sessions.

Phase-aware eviction. Planning and execution have different working set characteristics. Planning requires broad simultaneous context (many files held at once); execution is sequential (read, edit, move on). The proxy could detect phase from request patterns—many Reads with no Edits suggests planning—and adjust the eviction threshold accordingly.

Per-connection isolation. The current implementation uses a single PageStore for all connections. Subagent sessions share eviction state with the main session, causing cross-contamination. Per-connection isolation would give each session its own eviction history, fault tracking, and pinning state.

Trace-driven simulation. We extract reference strings (the sequence of page accesses) from session transcripts and replay them under different replacement policies. This enables evaluation of policies against recorded workloads without live inference-priced experiments. The inverted cost model (Section 6.2) means the optimal offline policy is not Belady's MIN but a cost-weighted variant that minimizes total keep cost plus fault cost. We plan to derive and evaluate this cost-optimal offline bound alongside MIN for comparison.

Cross-session access pattern prediction. The break-even formula requires estimating $T_{\text{until_next_ref}}$—the number of turns until a page is next referenced. Within a single session, this can only be estimated from local patterns. Across sessions, the prediction becomes tractable: coding sessions exhibit regular access patterns (orientation reads of project structure, iterative edits of a working file, periodic reference to configuration). A Markov model trained on cross-session reference strings could predict re-reference probability per page per turn, enabling cost-optimal eviction decisions. The data collection is nearly free: the proxy already logs every eviction, every fault, every tool call. What is needed is a persistent, queryable store of access patterns that survives session boundaries—precisely the role of a cross-session memory system.

Cooperative extensions. Phantom tools open a side-channel between the model and the proxy. The current vocabulary—release and fault—is minimal. Extensions include: *prefetch hints* (model signals upcoming working set needs), *eviction classes* (groups of pages evicted as a unit), *variable-fidelity tombstones* (AST skeletons instead of bare path markers), and *ephemeral flags* (tool results marked for immediate eviction after extraction). Each expands the model's ability to communicate cognitive state to the memory manager.

Beyond paging: object-addressed memory. The current system manages fixed-granularity pages addressed by file path. But the natural unit for LLM memory management is the semantic object—a conversation phase, a design decision, a debugging session. These objects vary in size, have relationships to each other, and can be compressed to multiple fidelity levels rather than the binary resident/evicted state of hardware pages. The eviction tombstone becomes a compressed summary that functions as both a page table entry (retrieval handle to the backing store) and a cache line (enough semantic content to answer queries without faulting). Declared losses in the summary tell the model what it *cannot* answer from the summary alone, guiding the decision to fault. The backing store is queryable—the model can ask a question of evicted content without materializing it in full. This shifts the abstraction from block-addressed paging to object-addressed memory management, where eviction is cooperative, compression is authored, and the backing store is a database rather than a swap partition.

Comparative measurement. Instrument other agentic tools (Cursor, Copilot, Windsurf, Continue.dev) to measure the same metrics. The structural argument predicts similar waste profiles; empirical confirmation would strengthen generalizability.

8 Conclusion

The context window of a large language model is not memory. It is L1 cache—a small, fast, expensive level in what should be a managed hierarchy. The field treats it as the entire memory system. There is no L2, no paging, no eviction policy, no working set estimation. The result: 21.8% structural waste in 4.45 billion measured tokens, with the model reprocessing stale content at 84.4× amplification.

Pichay demonstrates that the virtual memory abstractions developed for physical memory in the 1960s apply directly to LLM context windows. Demand paging with a simple FIFO policy recovers 36 percentage points of context capacity in steady-state use. Fault-driven pinning reduces repeated faults on working-set content in steady-state use. Retrieval handles—designed as space-saving markers—function as late-binding anchors that models understand without instruction. Graduated pressure zones and cooperative cleanup tags give

the model agency over its own memory management—a capability with no analogue in hardware memory hierarchies.

The system requires no changes to models, clients, or inference APIs. It is implemented as a transparent proxy and deployed in production use. The paper you are reading was written through it.

The deeper contribution is the architectural observation: LLM systems need not one large context window but a hierarchy of managed levels—from the generation window (L1) through persistent cross-session memory (L4)—connected by eviction policies and fault mechanisms. The first three levels are deployed; the first two are evaluated. L3—model-initiated conversation compaction—is the newest mechanism, and its evaluation is ongoing. The solutions for the remaining level exist in the OS literature. What remains is to connect them.

Authorship and AI Contribution

This paper was co-authored by a human systems researcher and an AI system (Claude Opus, Anthropic). The human contributed the memory hierarchy insight, research direction, experimental design, and editorial judgment—including the observation that context management is memory management, which reframed the entire project. The AI contributed code (analysis tools, proxy implementation, paging system), data analysis, and paper drafting.

Pichay was built by the system being studied—Claude Code running through its own proxy—and the paper was written through it. The evaluation data in Section 5.7 was generated as a side effect of the system's own development. We consider this self-referentiality a feature: the system's ability to measure, optimize, and sustain its own context usage over hundreds of turns is the strongest evidence that the interventions are practical.

All instruments and data are open source at: https://github.com/fsgeek/pichay. The paper snapshot cited in this manuscript is release `v0.1.0-paper` at commit `b56701a`. An immutable archival snapshot is available via Zenodo DOI: https://doi.org/10.5281/zenodo.18930122.

A Data Snapshot and Counting Protocol

To keep counts reproducible as the live corpora evolve, we separate paper numbers (frozen cohort for this draft) from live recounts. The counting protocol is versioned in the repository: `docs/corpus_protocol.md`.

Counting tool. Session counting is performed by `tools/corpus_counts.py`, which classifies Claude Code JSONL files into `main`, `subagent`, `compact`, `prompt_suggestion`, and `other`, and reports both raw and content-hash-deduplicated totals.

Recount command.

```
./tools/reproduce_paper_counts.sh
```

This writes a timestamped snapshot JSON to `paper/data/corpus_snapshot_YYYYMMDD_live.json`.

5-minute reproduction path. From repository root:

```
./tools/reproduce_paper_counts.sh
python3 tools/check_paper_numbers.py \
  --paper paper/main.tex \
  --snapshot paper/data/corpus_snapshot_20260307_live.json
latexmk -pdf paper/main.tex
```

Expected outputs: (1) a new snapshot JSON under `paper/data/`, (2) a paper-number consistency report, and (3) `paper/main.pdf`. On a typical laptop this path completes in minutes; the recount step is I/O-bound on local corpus size.

Current live recount artifact (2026-03-07). `paper/data/corpus_sna`
with roots: `~/.claude/projects` and `~/projects/yanantin/tmp/ubuntu-`
size filter `min_size=10000`, and Claude Desktop summary
from `~/projects/yanantin/tmp/claude-desktop/conversations.jso`

For this paper version, headline results use a frozen cohort to avoid drift across revisions; live recount artifacts are provided for auditability.

References

[1] Laszlo A. Belady. 1966. A study of replacement algorithms for a virtual-storage computer. *IBM Systems journal* 5, 2 (1966), 78–101.

[2] deepset Team. 2026. Context Engineering: The Next Frontier Beyond Prompt Engineering. https://www.deepset.ai/blog/context-engineering-the-next-frontier-beyond-prompt-engineering. Accessed March 2026.

[3] Peter J Denning. 1968. The working set model for program behavior. *Commun. ACM* 11, 5 (1968), 323–333.

[4] Peter J Denning. 1970. Virtual memory. *ACM Computing Surveys (CSUR)* 2, 3 (1970), 153–189.

[5] Yufeng Du, Minyang Tian, Srikanth Ronanki, Subendhu Rongali, Sravan Bodapati, Aram Galstyan, Azton Wells, Roy Schwartz, Eliu A Huerta, and Hao Peng. 2025. Context Length Alone Hurts LLM Performance Despite Perfect Retrieval. arXiv:2510.05381 [cs.CL] https://arxiv.org/abs/2510.05381

[6] Huiqiang Jiang, Qianhui Wu, Xufang Luo, Dongsheng Li, Chin-Yew Lin, Yuqing Yang, and Lili Qiu. 2024. Longllmlingua: Accelerating and enhancing llms in long context scenarios via prompt compression. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 1658–1677.

[7] Minki Kang, Wei-Ning Chen, Dongge Han, Huseyin A. Inan, Lukas Wutschitz, Yanzhi Chen, Robert Sim, and Saravan Rajmohan. 2025. ACON: Optimizing Context Compression for Long-horizon LLM Agents. arXiv:2510.00615 [cs.AI] https://arxiv.org/abs/2510.00615

[8] Sanjay Kariyappa and G. Edward Suh. 2026. SideQuest: Model-Driven KV Cache Management for Long-Horizon Agentic Reasoning. arXiv:2602.22603 [cs.AI] https://arxiv.org/abs/2602.22603

[9] T Kilbum, DBG Edwards, MJ Lanigan, and FH Sumner. [n. d.]. One-level storage system. In *IRE Transactions, EC-Denning, Peter J."On Modeling Program Behavior," Arlington, VA: AFIPS Press: Proceedings, Spring Joint Computer Conference*, Vol. 40. 937–944.

[10] Tobias Lindenbauer, Igor Slinko, Ludwig Felder, Egor Bogomolov, and Yaroslav Zharov. 2025. The Complexity Trap: Simple Observation Masking Is as Efficient as LLM Summarization for Agent Context Management. arXiv:2508.21433 [cs.SE] https://arxiv.org/abs/2508.21433

[11] Lingrui Mei, Jiayu Yao, Yuyao Ge, Yiwei Wang, Baolong Bi, Yujun Cai, Jiazhi Liu, Mingyu Li, Zhong-Zhi Li, Duzhen Zhang, Chenlin Zhou, Jiayi Mao, Tianze Xia, Jiafeng Guo, and Shenghua Liu.

2025. A Survey of Context Engineering for Large Language Models. arXiv:2507.13334 [cs.CL] https://arxiv.org/abs/2507.13334

[12] Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. 2024. MemGPT: Towards LLMs as Operating Systems. arXiv:2310.08560 [cs.AI] https://arxiv.org/abs/2310.08560

[13] Prithvi Rajasekaran, Ethan Dixon, Carly Ryan, and Jeremy Hadfield. 2025. Effective Context Engineering for AI Agents. https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents. Accessed March 2026.

[14] Cosmo Santoni. 2026. Contextual Memory Virtualisation: DAG-Based State Management and Structurally Lossless Trimming for LLM Agents. arXiv:2602.22402 [cs.SE] https://arxiv.org/abs/2602.22402

[15] Nikhil Verma. 2026. Active Context Compression: Autonomous Memory Management in LLM Agents. arXiv:2601.07190 [cs.AI] https://arxiv.org/abs/2601.07190

[16] Yuhang Wang, Yuling Shi, Mo Yang, Rongrui Zhang, Shilin He, Heng Lian, Yuting Chen, Siyu Ye, Kai Cai, and Xiaodong Gu. 2026. SWE-Pruner: Self-Adaptive Context Pruning for Coding Agents. arXiv:2601.16746 [cs.SE] https://arxiv.org/abs/2601.16746

[17] Simon Willison. 2025. Context Engineering. https://simonwillison.net/2025/jun/27/context-engineering/. Accessed March 2026.

[18] Philip Zeyliger. 2026. Expensively Quadratic: the LLM Agent Cost Curve. https://blog.exe.dev/expensively-quadratic. Accessed March 2026.